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2. (a) Sulfuric acid is produced by the contact process. The flow diagram shows the main stages in the process.

Stage **1**		Stage **2**		Stage **3**		Stage **4**
Burn raw material **A** in air to produce sulfur dioxide	→	Pass sulfur dioxide over vanadium(V) oxide at 450 °C to produce sulfur trioxide	→	Dissolve sulfur trioxide in concentrated sulfuric acid to produce oleum	→	Add a small amount of raw material **B** to oleum to produce concentrated sulfuric acid

(i) Give the name of raw materials **A** and **B**. [2]

A ....

B ....

(ii) State the purpose of vanadium(V) oxide in stage **2**. [1]

....

(iii) Complete and balance the equation for the reaction in stage **2**. [2]

$$\boxed{}\ \ldots\ldots\ldots\ldots + O_2 \rightleftharpoons 2SO_3$$

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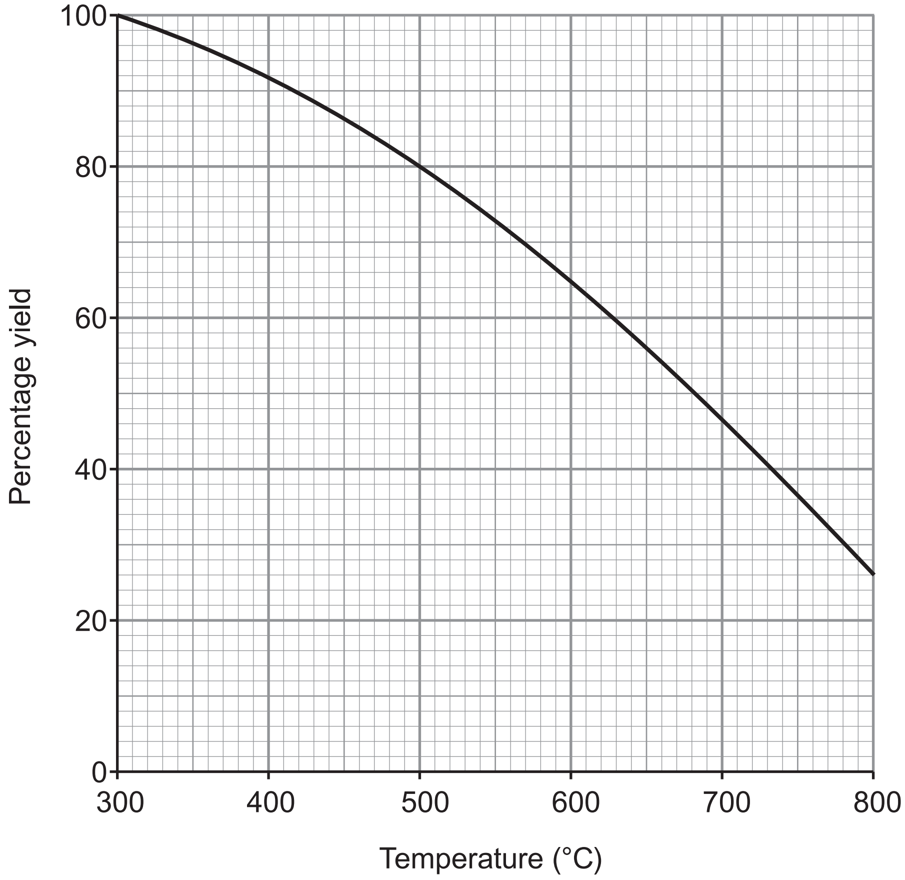
(iv) The graph shows how the percentage yield of sulfur trioxide changes with temperature between 300 °C and 800 °C.



Use the graph to answer parts I and II.

I. State the trend in the percentage yield of sulfur trioxide as the temperature increases. [1]

....

II. Give the temperature **range** to be used to obtain a yield of sulfur trioxide greater than 80 %. [1]

.... to .... °C

(v) One molecule of sulfur trioxide reacts with one molecule of sulfuric acid to form one molecule of oleum as the **only product**.

Complete the equation for this reaction by giving the formula of oleum. [1]

$SO_3 + H_2SO_4 \longrightarrow$ ....

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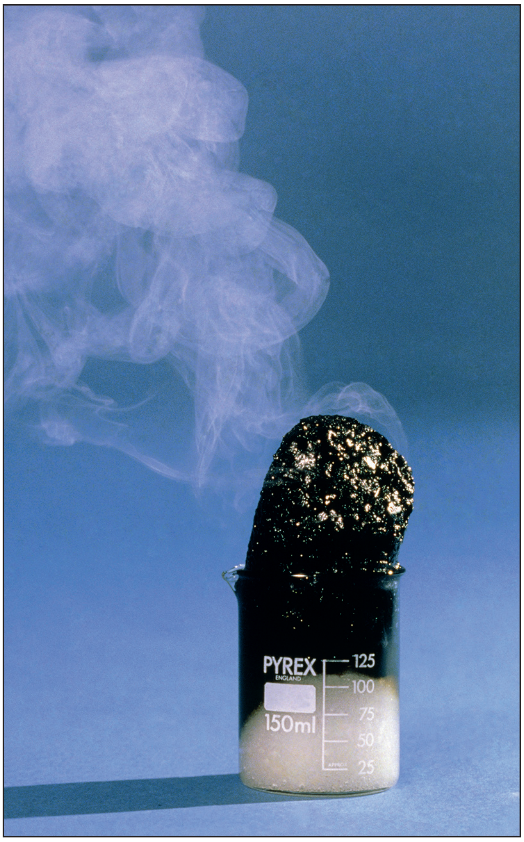
(b) The photograph shows the exothermic reaction between concentrated sulfuric acid and sugar, $C_{12}H_{22}O_{11}$.



(i) Name the **two** products formed. [2]

.... and ....

(ii) State the type of reaction taking place. [1]

....

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